Lab 7 & Quiz 7 Study Guide

Bridging the Control Plane and the Link Layer

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IT 213: Networking

Winter 2026

How to Prepare for the Quiz

- This presentation contains guiding concepts for your upcoming quiz.
- The answers are **not** explicitly given here.
- You must use the referenced sections in **Kurose & Ross (8th Edition)** to find the specific answers before taking the Moodle quiz.
- **Topics Covered:**
  - The Control Plane: Routing Algorithms & SDN (Chapter 5)
  - ICMP: Ping & Traceroute (Chapter 5)
  - The Link Layer: Framing, MAC, & Error Detection (Chapter 6)
  - Ethernet & Switches (Chapter 6)

Quiz Hints: The Control Plane (Ch. 5)

Routing Algorithms & SDN

- **Global vs. Decentralized:** What is the primary difference between a Link-State (LS) algorithm and a Distance-Vector (DV) algorithm regarding the network map they use? *(Read Section 5.2)*
- **OSPF:** Which of the two algorithm types mentioned above does OSPF actually use? *(Read Section 5.3)*
- **SDN Architecture:** In Software-Defined Networking, who or what actually computes the routing paths and installs the forwarding tables into the physical routers? *(Read Section 5.5)*

Quiz Hints: ICMP and Traceroute (Ch. 5)

Network Diagnostics

- **Traceroute Probing:** When you run a traceroute, what specific IPv4 header field is systematically modified to discover the routers along the path? *(Read Section 5.6)*
- **Error Messages:** When a router drops a traceroute probe, what specific ICMP *Type* and *Code* numbers are sent back as a warning? *(Read Section 5.6)*
- **Identifying the Drop:** How does the sender's computer know exactly which probe packet caused the router to generate the ICMP error? Look closely at what is included in the error payload. *(Read Section 5.6)*

Quiz Hints: Link Layer Basics (Ch. 6)

Moving across the physical wire

- **Core Responsibility:** While the Network layer handles end-to-end delivery, what is the exact physical scope of the Link Layer's responsibility? *(Read Section 6.1)*
- **Encapsulation:** What specific term is used to describe the layer-2 packet that is created when the Link Layer encapsulates a network-layer datagram? *(Read Section 6.1)*
- **Why add headers?:** What is the fundamental purpose of adding these link-layer headers and trailers before sending data over the physical wire? *(Read Section 6.1)*

Quiz Hints: Error Detection & MAC (Ch. 6)

Protecting the Data and Sharing the Channel

- **Error Codes:** Physical signals degrade. Which highly robust Error-Detecting Code standard is widely used in Ethernet and Wi-Fi? *(Read Section 6.2)*
- **The Cocktail Party:** What exactly is the "multiple access problem" when multiple nodes share a single broadcast link? What happens to their signals? *(Read Section 6.3)*
- **CSMA/CD:** Ethernet uses CSMA/CD. Understand the "listen before speaking" rule and what a node does if a collision is detected. *(Read Section 6.3)*

Quiz Hints: Ethernet & Addressing (Ch. 6)

MAC Addresses and Frame Structure

- **MAC vs. IP:** Understand the difference between an IP address (hierarchical, changes when moving) and a MAC address. What kind of structure does a MAC address have? *(Read Section 6.4)*
- **The Preamble:** What is the specific purpose of the 8-byte Preamble at the very beginning of an Ethernet frame? *(Read Section 6.4)*
- **Reliability:** Is standard Ethernet a reliable, connection-oriented service, or an unreliable, connectionless service? What happens to a frame that fails its CRC check? *(Read Section 6.4)*

Quiz Hints: ARP and Switches (Ch. 6)

Connecting the LAN

- **ARP:** If a host knows the destination IP address on its local network, which protocol translates it to find the unknown MAC address? *(Read Section 6.4)*
- **ARP Delivery:** When an ARP Request is generated, what special destination MAC address is used so that every node on the LAN receives it? *(Read Section 6.4)*
- **Switches vs. Routers:** What is the fundamental difference in the type of addresses (IP vs MAC) used by switches compared to routers when forwarding data? *(Read Section 6.4)*
- **Switch Tables:** Switches are "self-learning." How exactly do they populate their switch tables without administrator intervention? What happens if a destination is unknown? *(Read Section 6.4)*